

Comparing Penalized VAR Approaches for Intensive Longitudinal Data

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1 ABSTRACT

Estimating and forecasting multivariate time series from multiple individuals is challenging when each person's dynamics are partly shared and partly unique. In this project, we compare three approaches, LASSO VAR, standard Multi-VAR, and adaptive Multi-VAR and we test them on simulated multi-subject datasets of varying size, dimensionality, and lag structure. LASSO VAR models each individual separately and performs well for smaller, simpler datasets but struggles as dimensionality or lag order increases. Standard Multi-VAR jointly estimates shared and individual-specific effects, achieving the best overall balance of estimation error, sensitivity, specificity, and forecast accuracy in moderate-sized settings. Adaptive Multi-VAR improves specificity by more aggressively shrinking small coefficients, but sometimes misses true relationships, reducing sensitivity. Our results demonstrate trade-offs between accurately recovering transition structures and maintaining forecast performance. These findings provide information for choosing estimation strategies for intensive longitudinal data, highlighting multi-individual models when analyzing more complex or real-world datasets.

2 INTRODUCTION

This project examines how to estimate and forecast multivariate time series collected from multiple individuals when their dynamics are partly shared and partly person-specific. We study this using three approaches: individual LASSO VAR, standard multi-VAR, and adaptive multi-VAR.

This problem is especially relevant for intensive longitudinal data (ILD) in areas such as psychology, neuroscience, public health, and mobile-sensing research. These data sets are often high-dimensional and noisy, and understanding how much structure is common across people versus truly individual helps researchers better describe the processes

driving behavior. Stronger estimation naturally leads to better forecasting, which matters for predicting real-time emotional or physiological states.

Distinguishing shared structure from individual-specific effects also improves forecasts when each subject’s time series is short or noisy. Accurate predictions can support clinical or behavioral decisions by signaling when a state shift may occur and who may need an intervention.

In this project, we compare LASSO VAR models, for both VAR(1) and VAR(2), with the standard and adaptive multi-VAR frameworks, which assume a VAR(1) structure and pool information across subjects. We evaluate all methods using simulated multi-subject data and assess structure recovery (estimation error, sensitivity and specificity), and forecasting accuracy (RMSFE).

The rest of the report is organized as follows. We first describe the generative model and the simulation setup, then outline each estimation procedure. Next, we present results on structure recovery and forecasting, and conclude with a comparison across all approaches, including a contrast between VAR(1) and VAR(2) for the LASSO methods.

Throughout, we focus on three central questions: (1) how to jointly model multiple individuals while still allowing person-specific effects; (2) how penalized VAR methods improve estimation in high dimensions; and (3) whether multi-VAR approaches improve forecasting compared to single-subject VARs.

3 METHODOLOGY

Consider a d -dimensional multivariate time series $\{\mathbf{X}_t\}_{t \in \mathbb{Z}}$, where $\mathbf{X}_t = (X_{1,t}, \dots, X_{d,t})'$. For a fixed lag order $p \geq 1$, a vector autoregressive model of order p , denoted VAR(p), is given by

$$\mathbf{X}_t = \Phi_1 \mathbf{X}_{t-1} + \dots + \Phi_p \mathbf{X}_{t-p} + \varepsilon_t, \quad t \in \mathbb{Z}, \quad (3.1)$$

where $\Phi_1, \dots, \Phi_p \in \mathbb{R}^{d \times d}$ are coefficient matrices and $\varepsilon_t \sim N(0, \Sigma_\varepsilon)$ are independent across t . A unique causal stationary solution to (3.1) exists if $\det(\Phi(z)) \neq 0$ for all complex z with $|z| \leq 1$, where $\Phi(z) = I_d - \Phi_1 z - \dots - \Phi_p z^p$.

If we directly use ordinary least squares (OLS) to estimate a VAR model, the estimator has good theoretical properties; for example, it is equivalent to the maximum likelihood estimator under normal errors. However, in high-dimensional settings OLS can overfit, produce spurious relationships, and lead to an ill-conditioned design matrix $\mathbf{X}'\mathbf{X}$.

3.1 ESTIMATION OF SPARSE VAR MODELS

Given a sample $\mathbf{X}_1, \dots, \mathbf{X}_T$, we focus on estimating the coefficient matrices $\Phi_1, \dots, \Phi_p$ in a high-dimensional setting where d and p can be large. Let $N = T - p$ and stack the data in regression form as

$$\mathbf{Y} = \begin{pmatrix} \mathbf{X}'_{p+1} \\ \mathbf{X}'_{p+2} \\ \vdots \\ \mathbf{X}'_T \end{pmatrix} \in \mathbb{R}^{N \times d}, \quad \mathbf{Z} = \begin{pmatrix} \mathbf{X}'_p & \mathbf{X}'_{p-1} & \dots & \mathbf{X}'_1 \\ \mathbf{X}'_{p+1} & \mathbf{X}'_p & \dots & \mathbf{X}'_2 \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{X}'_{T-1} & \mathbf{X}'_{T-2} & \dots & \mathbf{X}'_{T-p} \end{pmatrix} \in \mathbb{R}^{N \times pd},$$

and

$$\mathbf{B}^* = \begin{pmatrix} \Phi'_1 \\ \Phi'_2 \\ \vdots \\ \Phi'_p \end{pmatrix} \in \mathbb{R}^{pd \times d}, \quad \mathbf{E} = \begin{pmatrix} \varepsilon'_{p+1} \\ \varepsilon'_{p+2} \\ \vdots \\ \varepsilon'_T \end{pmatrix} \in \mathbb{R}^{N \times d}.$$

Then the VAR(p) model can be written as

$$\mathbf{Y} = \mathbf{Z}\mathbf{B}^* + \mathbf{E}, \quad (3.2)$$

so that estimating the VAR coefficients is equivalent to fitting a multivariate linear regression model with response matrix $\mathbf{Y}$, design matrix $\mathbf{Z}$, and coefficient matrix $\mathbf{B}^*$.

In high-dimensional settings, the total number of parameters pd^2 can be very large. To avoid overfitting and to improve interpretability, we assume that many coefficients are zero and impose sparsity on $\mathbf{B}^*$, using penalized estimators such as LASSO-type methods.

3.2 METHOD (A): INDIVIDUAL LASSO VAR

Assume that the coefficient matrix $\mathbf{B}^*$ is sparse, in the sense that only a small number of its entries are nonzero. The LASSO estimator $\hat{\mathbf{B}}$ is obtained by minimizing the average squared error plus an ℓ_1 penalty on the coefficients:

$$\hat{\mathbf{B}} = \arg \min_{\mathbf{B} \in \mathbb{R}^{pd \times d}} \left\{ \frac{1}{N} \|\mathbf{Y} - \mathbf{Z}\mathbf{B}\|_F^2 + \lambda \|\mathbf{B}\|_1 \right\}, \quad (3.3)$$

where $\|\cdot\|_F$ denotes the Frobenius norm and $\|\mathbf{B}\|_1 = \sum_{j,k} |B_{jk}|$ is the ℓ_1 -norm penalty. The tuning parameter $\lambda > 0$ controls the strength of the penalty and hence the degree of sparsity. Equation (3.3) is equivalent to solving d separate LASSO regressions, one for each component time series.

3.3 METHOD (B): THE MULTI-VAR APPROACH

Suppose now that we have K individuals, and each one has its own VAR coefficient matrix $\mathbf{B}_k^*$. To borrow strength across individuals, we decompose $\mathbf{B}_k^*$ as

$$\mathbf{B}_k^* = \boldsymbol{\mu}^* + \boldsymbol{\Delta}_k^*, \quad k = 1, \dots, K,$$

where $\boldsymbol{\mu}^* \in \mathbb{R}^{pd \times d}$ represents the effects that are common across all individuals, and $\boldsymbol{\Delta}_k^* \in \mathbb{R}^{pd \times d}$ captures the subject-specific deviations for individual k .

The multi-VAR LASSO estimator is then defined as

$$(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Delta}}_1, \dots, \hat{\boldsymbol{\Delta}}_K) = \arg \min_{\boldsymbol{\mu}, \boldsymbol{\Delta}_1, \dots, \boldsymbol{\Delta}_K} \left\{ \frac{1}{N} \sum_{k=1}^K \|\mathbf{Y}^{(k)} - \mathbf{Z}^{(k)}(\boldsymbol{\mu} + \boldsymbol{\Delta}_k)\|_F^2 + \lambda_1 \|\boldsymbol{\mu}\|_1 + \sum_{k=1}^K \lambda_{2,k} \|\boldsymbol{\Delta}_k\|_1 \right\}. \quad (3.4)$$

The first penalty, with tuning parameter λ_1 , encourages sparsity in the common effect $\boldsymbol{\mu}$. The second set of penalties $\{\lambda_{2,k}\}_{k=1}^K$ shrinks the individual deviations $\boldsymbol{\Delta}_k$. When heterogeneity across individuals is high, the multi-VAR estimator behaves similarly to fitting K separate LASSO VAR models.

3.4 METHOD (C): THE ADAPTIVE MULTI-VAR APPROACH

To improve variable selection and reduce the bias of the LASSO estimator, we further consider an adaptive version of the multi-VAR estimator. Let $\tilde{\mathbf{B}}$ be a pilot estimator of the common coefficient matrix (for example, obtained from (3.3) or (3.4)), and define positive weight vectors $\boldsymbol{\omega} = (\omega_j)$ and $\mathbf{v}_k = (v_{k,j})$ by

$$\omega_j = \frac{1}{|\tilde{B}_{\cdot,j}|}, \quad v_{k,j} = \frac{1}{|\tilde{B}_{k,j} - \tilde{B}_{\cdot,j}|},$$

where $\tilde{\mathbf{B}}_k$ is a preliminary estimate for individual k and $|\cdot|$ denotes the absolute value, with a small ridge offset added in the denominator in practice.

The adaptive multi-VAR estimator is obtained by minimizing the objective function

$$\frac{1}{N} \sum_{k=1}^K \|\mathbf{Y}^{(k)} - \mathbf{Z}^{(k)}(\boldsymbol{\mu} + \boldsymbol{\Delta}^{(k)})\|_F^2 + \lambda_1 \left(\sum_j \omega_j |\mu_j| + \sum_{k=1}^K \frac{\lambda_{2,k}}{\lambda_1} \sum_j v_{k,j} |\Delta_j^{(k)}| \right), \quad (3.5)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Delta}^{(1)}, \dots, \boldsymbol{\Delta}^{(K)}$ are the common and subject-specific coefficient matrices to be estimated. The adaptive weights penalize small pilot coefficients more heavily and large pilot coefficients less heavily, which reduces shrinkage bias on truly nonzero entries.

Under suitable regularity conditions, adaptive LASSO-type estimators have the oracle property: they correctly identify the set of nonzero coefficients with high probability and achieve asymptotically efficient estimation of the nonzero entries. By combining this idea with the multi-individual structure, the adaptive multi-VAR estimator in (3.5) aims to provide more accurate and interpretable estimates of both the common and individual-specific VAR effects.

4 IMPLEMENTATION

We first implemented code to generate stable VAR coefficient matrices. After drawing candidate transition matrices, we enforced stability by rescaling them when needed so that the associated companion matrix had spectral radius less than one. To generate time series data, we used a burn-in period of $T_{\text{burn}} = 50$, discarding the first 50 observations and keeping the remaining T points for analysis.

For the multi-VAR approach, we utilized the R package `multivar`, which implements a joint penalized estimator for multi-individual VAR models. In our study, we used blocked cross-validation (`cv = "blocked"`, `nfolds = 10`) to tune the penalty parameters. The main implementation challenge was selecting suitable ranges for the two tuning parameters λ_1 and λ_2 across all simulation designs; in the final analysis we relied on the default grids provided by `multivar`, as manual tuning did not yield a clearly superior choice.

The individual LASSO VAR baseline was implemented in R using the `glmnet` package. For each subject, we stacked the VAR data into a standard regression form by constructing a response matrix of current observations and a design matrix of lagged predictors, and then applied `cv.glmnet` with an ℓ_1 penalty.

5 DATA ANALYSIS

5.1 SIMULATION STUDY

To evaluate the performance of the proposed penalized VAR methods for forecasting multi-subject ILD, we conducted a simulation study with 16 settings. We varied the number of individuals $K \in \{10, 20\}$, the time series length per individual $T \in \{30, 50\}$, the number of variables per subject $d \in \{10, 20\}$, and the VAR lag order $p \in \{1, 2\}$.

Across all settings, the coefficient matrices were generated to be sparse, with 5% of entries nonzero; for instance, when $d = 20$ this corresponds to 20 nonzero coefficients in each $d \times d$ transition matrix. We fixed the heterogeneity at the medium level so that each individual shared a common VAR structure with moderate subject-specific deviations. Nonzero coefficients were drawn from $\text{Uniform}(0.1, 0.9)$ and rescaled, when necessary, to satisfy the VAR stability condition. White noise was added with standard deviation $\sigma_\varepsilon = 0.1$. For forecasting evaluation, we held out the last five time points for each individual and used the remaining data for model fitting.

K	T	d	Setting No.	Setting No.
			$(p = 1)$	$(p = 2)$
10	30	10	1	2
10	30	20	3	4
10	50	10	5	6
10	50	20	7	8
20	30	10	9	10
20	30	20	11	12
20	50	10	13	14
20	50	20	15	16

Table 5.1: Simulation designs

5.2 OUTCOME MEASURES

We evaluate the performance of each method using four outcome measures: estimation error, mean sensitivity, mean specificity, and mean root mean squared forecast error (RMSFE).

Let $\mathbf{B}_k^*$ and $\hat{\mathbf{B}}_k$ denote the true and estimated transition matrices for individual k , respectively. The estimation error summarizes how close the estimated coefficients are to the truth,

$$\text{Estimation Error} = \frac{1}{K} \sum_{k=1}^K \|\hat{\mathbf{B}}_k - \mathbf{B}_k^*\|_F. \quad (5.1)$$

Mean sensitivity and mean specificity measure variable-selection performance:

$$\text{Mean Sensitivity} = \frac{1}{K} \sum_{k=1}^K \left(\frac{\sum_j \mathbf{1}\{\hat{B}_{k,j} \neq 0 \text{ and } B_{k,j}^* \neq 0\}}{\sum_j \mathbf{1}\{B_{k,j}^* \neq 0\}} \right), \quad (5.2)$$

$$\text{Mean Specificity} = \frac{1}{K} \sum_{k=1}^K \left(\frac{\sum_j \mathbf{1}\{\hat{B}_{k,j} = 0 \text{ and } B_{k,j}^* = 0\}}{\sum_j \mathbf{1}\{B_{k,j}^* = 0\}} \right), \quad (5.3)$$

where $\mathbf{1}\{\cdot\}$ denotes the indicator function. Finally, forecasting performance is assessed using the mean root mean squared forecast error (RMSFE). Let n_{hold} be the number of holdout time points and $Y_{j,t}^{(k)}$, $\hat{Y}_{j,t}^{(k)}$ the true and predicted values for variable j of individual k at time t . Then

$$\text{Mean RMSFE} = \frac{1}{K} \sum_{k=1}^K \sqrt{\frac{1}{d n_{\text{hold}}} \sum_{j=1}^d \sum_{t \in \mathcal{T}_{\text{hold}}} (\hat{Y}_{j,t}^{(k)} - Y_{j,t}^{(k)})^2}, \quad (5.4)$$

where $\mathcal{T}_{\text{hold}}$ indexes the holdout time points for forecast evaluation.

5.2.1 ESTIMATION ERROR

For estimation error, LASSO (Lag 1) works well only in low to moderate dimensional settings, but its error increases quickly as K and D grow. LASSO (Lag 2) follows the same pattern and generally performs slightly worse, especially in high-dimensional scenarios where its errors become the largest overall. Multi-VAR (Standard) consistently achieves the lowest estimation error across almost all settings and degrades more slowly as complexity increases. Multi-VAR (Adaptive) improves upon LASSO in many cases but does not outperform the standard Multi-VAR and often overshinks coefficients when dimensions increase, causing its error to rise again.

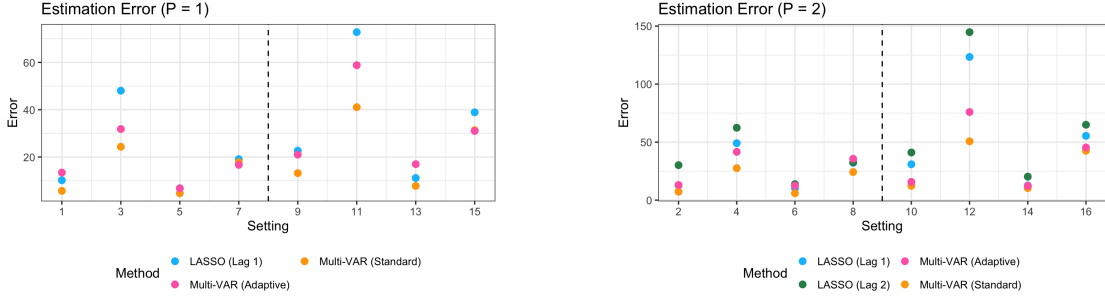


Figure 5.1: Estimation Error Simulation Means

5.2.2 SENSITIVITY

For sensitivity, LASSO (Lag 1) and LASSO (Lag 2) show moderate recovery of true signals, but their performance drops once the system becomes high-dimensional, as seen

in the lag-2 results when $K=20$. Multi-VAR (Standard) achieves the strongest and most stable sensitivity across settings, consistently identifying true nonzero entries even as K and D increase. Multi-VAR (Adaptive) has the weakest sensitivity overall, since its aggressive shrinkage often removes true effects, especially in more complex scenarios. This pattern is most noticeable in smaller-parameter simulations; for example, in the Lag 1 plot, Settings 2 and 6 show very low sensitivity despite only having $K=10$ and $D=10$.

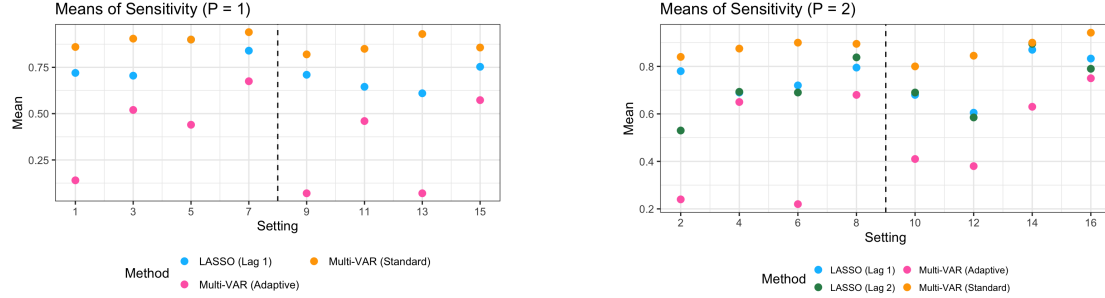


Figure 5.2: Sensitivity Simulation Means

5.2.3 SPECIFICITY

Specificity moves in the opposite direction of sensitivity. LASSO (Lag 1) and LASSO (Lag 2) keep a solid level of specificity across most settings, and Multi-VAR Adaptive pushes this even higher, often reaching values close to one because its heavier penalization removes almost all small coefficients. This contrast is especially noticeable given how weak its sensitivity is. Meanwhile, Multi-VAR Standard consistently shows the lowest specificity, since the joint estimation tends to include more coefficients and increases the likelihood of false positives when the dimension grows.

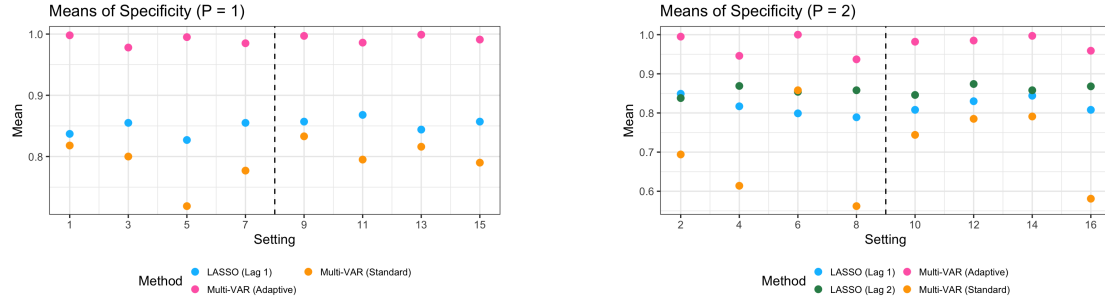


Figure 5.3: Specificity Simulation Means

5.2.4 RMSFE

For forecast accuracy, LASSO (Lag 1) and LASSO (Lag 2) stay the most stable, with RMSFE values hovering around 0.10–0.15 even as the dimension increases. In the first

plot for $p=1$ that might look questionable at first, but once you pay attention to the y-axis, the differences are actually very small. Multi-VAR (Standard) holds up in the moderate settings but becomes unstable in the most difficult ones, where its RMSFE jumps noticeably. Multi-VAR (Adaptive) shows a similar pattern: solid accuracy when the system is manageable, but large spikes when the dimension grows or the structure gets more complicated. In the second plot for $p=2$, settings 4 and 8 stand out with sharp increases in RMSFE, which makes sense because those runs use a smaller $K=10$ paired with a larger $d=20$ and higher lag order $p=2$, making the forecasting step much harder.

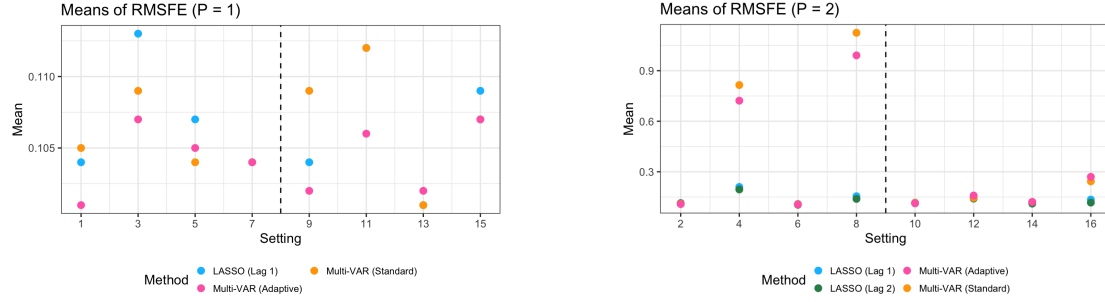


Figure 5.4: RMSFE Simulation Means

5.2.5 KEY RESULTS

These findings help answer the key questions from the introduction. First, Multi-VAR (Standard) provides the strongest evidence that multiple individuals can be modeled jointly while still allowing person-specific effects, since it consistently achieves the lowest estimation error and the highest sensitivity across most settings. Second, the LASSO-based methods show how penalized VAR models help in high-dimensional systems: they stabilize estimation, reduce overfitting, and maintain reasonable specificity even as K , D , and p increase. Finally, multi-VAR approaches do not necessarily improve forecasting compared to single-subject models. The RMSFE results indicate that all methods perform similarly for $p = 1$, but when $p = 2$, the LASSO models remain the most stable and deliver the most reliable out-of-sample forecasts.

6 CONCLUSIONS

The results from our analyses show that Multi-VAR Standard is the most effective approach for recovering shared and individual-specific dynamics in smaller dimensional settings. Across all sixteen simulations, it consistently produced low estimation error and high sensitivity, indicating that it captures true transitions well. Its specificity was more limited, suggesting a tendency to include extra coefficients, but overall it offered the strongest balance for structure recovery.

Multi-VAR Adaptive showed the opposite trade-off: high specificity but weaker sensitivity. It avoided false positives but often removed true effects, especially as dimensionality

increased. This pattern suggests that future work could focus on improving penalty structures or tuning strategies that preserve important signals while maintaining sparsity.

All methods performed best in low to moderate dimensions, and their estimation accuracy and forecasting performance declined as K , D , and p increased. This trend highlights the difficulty of high-dimensional VAR estimation and suggests that larger sample sizes or more flexible VAR(p) structures may be necessary when dynamics are more complex.

Expanding the simulations to larger parameter settings remains an important direction. Running models with more individuals, more variables, longer time series, and higher lag orders would clarify how well these methods scale and how stable they remain in truly high-dimensional or heavily dynamic systems. Another key direction is developing approaches that better balance sensitivity and specificity, particularly for the Adaptive Multi-VAR model, which showed strong sparsity but at the cost of missing true effects. Finally, applying these models to real intensive longitudinal datasets would provide insight into practical performance outside of controlled simulations and reveal how well these methods capture real-world temporal structure.

7 REFERENCES

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8 TEAM MEMBER CONTRIBUTION

Nancy Lopez contributed the abstract, introduction, data analysis, and conclusion sections of the report and slide show presentation. She also implemented approximately half of the project’s code and produced visualizations of the results.

Pei-Ling Lin contributed the methodology and implementation sections of the report. She also developed the remaining portion of the project’s code and generated additional visualizations related to the methodological procedures.